Examiner only

Answer **all** questions.

1. (a) Plant leaves absorb light for photosynthesis.

State the source of the light absorbed by the leaves. [1]

....

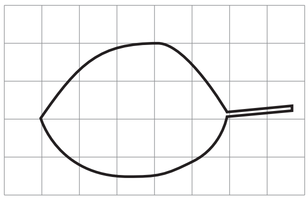
(b) **Image 1.1** shows a violet plant (*Viola riviniana*).

Image 1.1

Gareth calculated the area of a leaf from one of the violet plants. This is what he did:

Step 1. He drew an outline of a leaf on a grid as shown in **Image 1.2**.

Image 1.2



Step 2. He counted the number of **whole** squares inside the outline.

Step 3. He multiplied the total number of whole squares by the area of one square.

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(i) Count the number of **whole** squares inside the outline. [1]

Number of **whole** squares inside the outline = ...

(ii) Each square has an area of $0.25\,cm^2$.
Use the formula below to calculate the outline area. [1]

Outline area = 0.25 × number of whole squares inside the outline.

Outline area = ... cm^2

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(iii) Counting only the whole squares gave an outline area which was much lower than the actual leaf area.

I. Explain why. [1]

....

II. Suggest **one** improvement to **step 2** to get an answer closer to the true value. [1]

....

....

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(iv) Gareth decided to test the following hypothesis:

"The leaf area of violet plants in the shade will be greater than that of violet plants in bright light".

He picked one violet plant growing in shade and one growing in bright light.

Gareth used his method to calculate the leaf areas of 10 leaves from each plant.

The mean results are shown in **Table 1.3**.

Table 1.3

Plant growing in	Mean leaf area (cm^2)
shade	1.5
bright light	0.9

I. Calculate the difference in mean leaf area between the two plants. [1]

Difference = ... cm^2

II. **Complete the following sentence** by underlining the correct word. [1]

The results show that Gareth's hypothesis is:

invalid / rejected / supported.

(c) Explain the advantage of a large leaf area for plants growing in shade. [2]

....

....

....

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(iii) At the start of the investigation, Tracy made two predictions:

1. The temperature of the peas in flask **A** will rise but the temperature of the peas in flask **B** will not change.

2. Eventually the temperature in flask **A** will stop rising.

Use the results to state whether each of Tracy's predictions were correct. Explain your answers. [2]

Prediction **1** ....

....

Prediction **2** ....

....

(iv) Suggest **one** practical way in which Tracy could confirm or reject her second prediction. [1]

....

....

(v) Tracy recorded the mass of the peas in flask **A** at the start and at the end of the investigation. **Complete the sentence below** by underlining the correct outcome that you would expect Tracy to observe. [1]

During the investigation, the mass of the peas in flask **A** will have:

decreased **increased** **remained the same**

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